

Demonstration of Proposed Features

Date	01 JULY 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrate d (Yes / No)	Remarks
1	Data Engineering	Cleans duplicates and programmatically derives binary target risk labels (TARGET=1 for delinquency overdue by 30+ days, TARGET=0 otherwise).	Implemented	Yes	Merged application and credit history tables, engineered targets, and handled duplicate IDs.
2	Exploratory Data Analysis (EDA)	Automatic visual plots of client demographics, income density distributions, default ratios, and correlation heatmaps.	Implemented	Yes	Generates 6 demographic distribution plots and correlation heatmaps saved to static/images/plots/.
3	Modular Preprocessing	Bundles numerical scaling and categorical encoding into a serialized Scikit-learn ColumnTransformer pipeline.	Implemented	Yes	Fitted pipeline is saved as models/preprocessor.joblib to prevent data leakage during web predictions.
4	Model Selection	Compares tuned Logistic Regression, Decision Trees, XGBoost, and Random Forests using cross-validation.	Implemented	Yes	Random Forest selected as champion with 80.44% accuracy, 0.3937 F1-score, and 0.7559 ROC-AUC.
5	Web Dashboard	Web application with a vertical sidebar to toggle between Application Form, Model Metrics, and EDA Plots	Implemented	Yes	Fully functional Flask dashboard with client-side validation and animated circular gauge risk meter.
6	Production Build Automation	Single-command build file (build_assets.py) that sets up databases and fits model weights on the server.	Implemented	Yes	Successfully downloads raw datasets, compiles features, fits preprocessing, and trains model in one command.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100